STUDENT PROJECTS LABORATORY - 930 KINNEAR RD

STANDARD OPERATING PROCEDURE (SOP) | ASSET: WLD-05 | TAG: THERM-BM-MF

WLD-05: Thermal Processing Suite: Blue M Industrial Oven & Muffle Furnace

Facility Zone: Zone 3: Welding & Thermal Processing (WLD) | Supervisor: Max Wright (Sr. Instructional Lab Supervisor)

1. REQUIRED PERSONAL PROTECTIVE EQUIPMENT (PPE)

1. ANSI Z87.1 Safety Goggles / Face Shield
2. High-Temperature Insulated Kevlar/Aluminized Gloves
3. Flame-Resistant (FR) Lab Coat & Closed-Toe Shoes

2. PRIMARY OPERATIONAL HAZARDS

1. Extreme thermal burn and radiant heat hazard (up to 650 deg F in Blue M, 1,250 deg C in Muffle Furnace)
2. Explosion/fire hazard if flammable solvents, resins, or volatiles are placed inside (units are NON-explosion-proof)
3. Thermal burping (superheated air blast) if Blue M door is opened above 200 deg F
4. Blower motor bearing seizure if Blue M power is shut off above 167 deg F (75 deg C)

3. BEFORE OPERATION CHECKLIST

1. Strictly NO flammable solvents, uncured volatiles, or sealed containers inside either chamber.
2. Set Over-Temperature Protection (OTP) controller 15 deg C above main process setpoint.

4. DURING OPERATION SAFETY RULES

1. Follow the '200 deg F Rule': Do NOT open the Blue M oven door when chamber temperature exceeds 200 deg F to prevent thermal burping.
2. Use long high-temperature tongs when loading/unloading Muffle Furnace.

5. AFTER USE & SHOWROOM CLEANUP

1. Mandatory Blue M Cool-Down: Lower setpoint below 167 deg F (75 deg C) and keep blower running until chamber drops below 167 deg F before switching power OFF.

EMERGENCY SHUTDOWN & INCIDENT PROTOCOL:

Shut off heating program, disconnect wall switch/plug if safe, keep chamber door CLOSED to contain any smoke/fire, and notify Lab Supervisor.